Academic & Technical Project Proposal

Anisotropy-Aware De Novo Structural Elucidation Engine (ADSEE)

1. Executive Summary & Objective

Modern Computer-Assisted Structure Elucidation (CASE) frameworks have increasingly drifted toward deep learning models, neural networks, and probabilistic architectures. While these methodologies show impressive results on standard molecular profiles, they suffer from two critical flaws: they are inherently opaque black-box systems, and they fail spectacularly when evaluating complex small-molecules characterized by strong magnetic anisotropic environments (e.g., highly folded structures containing multiple aromatic ring systems). This document describes the formal technical design of ADSEE (Anisotropy-Aware De Novo Structural Elucidation Engine), a highly optimized, lightweight, deterministic expert system designed to replicate an expert structural chemist's precise analytical deduction process.

The primary objective of ADSEE is to perform automated 'macro-filtering' and high-precision graph-topological triangulation to identify structural assignments of obscure small molecules (typically under 500 g/mol) generated via high-field multi-dimensional nuclear magnetic resonance (NMR) spectroscopy. By implementing strict physical logic gates, deterministic graph theory, and traditional chemical physics, the pipeline compresses infinite molecular configurations into a precise handful of verified structural solutions in less than a second on consumer-grade hardware. The final output is intended as a client-side serverless application, eliminating bloated infrastructure in favor of sheer computational efficiency.

2. Core Logic & Algorithmic Pipeline

Instead of trying to resolve the entire spectrum simultaneously—a task prone to exponential combinatoric explosions—ADSEE applies a strict hierarchical pruning strategy. Each subsequent stage increases computational precision as it operates over a heavily minimized pool of candidate molecules.

Stage 1: Exact Mass Filtration (MS Filter)

Methodology: The process initiates at the coarsest level of resolution by ingesting high-accuracy mass spectrometry data. Given an experimental monoisotopic mass obtained from a high-resolution spectrometer (e.g., Orbitrap or Q-TOF) with an accuracy of ±0.0005 Da, the program dynamically generates or matches the unique molecular formula constraint. A relational database query is immediately executed against a localized flat-file compound library (curated from open repositories like PubChem or ChEBI). This step strips away millions of structurally irrelevant compounds, yielding a highly constrained search space of approximately 100 to 500 structural isomers.

Stage 2: 1D Graph-Topological Filter (Spin System Checklist)

Methodology: At high magnetic field strengths (such as 800 MHz), 1D 1H spectra resolve into distinct, highly defined spin networks. Users specify macro-level structural components that are visually unmistakable (e.g., a symmetric para-substituted aromatic system, an isolated ethyl

chain, or a distinct tert-butyl singlet). ADSEE avoids the premature and computationally expensive route of exact chemical shift simulation. Instead, it treats the 1D spectrum as a discrete structural component checklist. The engine translates these macro spin-systems into highly precise SMARTS patterns and feeds them into a multi-threaded structural matching engine powered by RDKit. If a candidate isomer does not contain the literal graph connectivity required to manifest those spin patterns, it is instantaneously purged from the evaluation queue. This drops the candidate pool from several hundred down to a manageable array of 5 to 12 compounds.

Stage 3: Physical Anisotropy Shielding Correction

Methodology: Standard empirical chemical shift prediction software utilizes linear increment methodologies that assume isolated local chemical environments. These applications generate massive errors when molecules fold, forcing protons to reside directly above or within the shielding cone of a spatial aromatic ring, inducing severe unpredicted upfield or downfield shifts. ADSEE natively addresses this physical phenomenon. For the 5 to 12 surviving candidates, the engine executes a rapid 3D conformation generation sequence using the Merck Molecular Force Field (MMFF94). The program then applies the classical McConnell equation for magnetic anisotropy to the spatial coordinates of the 3D conformer:

$$\Delta\sigma = (\Delta\chi / (3 * r^3)) * (1 - 3*\cos^2\theta)$$

Where $\Delta\chi$ represents the magnetic susceptibility anisotropy of the ring system, r defines the literal Euclidean distance from the center of the ring to the targeted proton nucleus, and θ represents the angle relative to the ring's principal symmetry axis. By dynamically shifting empirical base predictions using this structural adjustment factor, ADSEE bypasses the pitfalls that blind commercial increment tools, filtering out mismatched regioisomers with flawless precision.

Stage 4: 2D Matrix Triangulation (The Ruler Method)

Methodology: The final phase automates the chemist's manual cross-referencing of multi-dimensional data arrays. The experimental 2D cross-peaks (extracted directly from standard HSQC and HMBC outputs) are read as a coordinate matrix of connectivity constraints. The engine executes a Subgraph Isomorphism algorithm. For each remaining molecular structure, it validates every single experimental peak against the candidate's actual bonding graph. Heteronuclear Single Quantum Coherence (HSQC) correlations are validated strictly against 1-bond pathways, while Heteronuclear Multiple Bond Correlation (HMBC) signals are checked against rigorous 2- to 3-bond paths. Structures requiring structural leaps beyond these physical graph boundaries are heavily penalized or discarded. This eliminates stereochemical or regional ambiguities, isolating the true molecular structure.

3. Target Minimalist Architecture

The core architectural mantra of ADSEE is absolute zero bloat, native speed, and ultimate desktop portability. To achieve this, the system avoids complex server deployments, cloud dependencies, or microservice environments.

- **Core Engine Language:** C++20 represents the foundation of the calculation engine. This choice ensures raw memory manipulation capability, blistering execution speeds during dense graph traversals, and deep integration with native high-performance libraries.
- **Cheminformatics Backend:** Rather than recreating chemical informatics infrastructure, ADSEE embeds the native C++ API of RDKit. This handles ultra-fast SMARTS evaluation, canonicalization of SMILES, and 3D geometric force-field coordinate computations natively.
- **Compilation Target:** To satisfy the desire for a highly interactive, lightning-fast web deployment without sacrificing native desktop execution speeds, the entire C++ codebase will be compiled into WebAssembly (Wasm) utilizing the Emscripten toolchain. This allows the complex graph algorithms and physics engines to run safely inside a client browser thread at near-native speeds, powered entirely by the user's local hardware as a static, serverless single-page app.
- **Data Storage:** Eschewing heavy relational database servers (e.g., PostgreSQL/MySQL), candidate molecular data catalogs will be loaded via structured, packed local flat-files (CSV layouts), ensuring the application can run in offline air-gapped environments without architectural degradation.

4. Algorithmic Data & Pipeline Blueprint Matrix

The matrix below cross-references each distinct pipeline stage with its corresponding computational complexity, data input types, and localized performance targets.

Pipeline Stage	Primary Input Type	Algorithmic Complexity	Target Execution Core
Stage 1: MS Filter	HRMS Monoisotopic Mass (Da)	O(N) Linear Scan	Multi-Threaded CPU Scan
Stage 2: 1D Graph Filter	User SMARTS Checklists	O(M * K) Graph Pattern	Parallelized CPU Matching
Stage 3: Anisotropy Engine	Conformer Spatial Coordinates	O(I * J) Point Calculations	CPU / SIMD Vector Loops
Stage 4: 2D Matrix Triangulation	HSQC / HMBC CSV Matrices	O(V + E) Shortest-Path Check	Client-Side Wasm Core

5. Phase-Gate Prototype Development Blueprint

Phase 1: Environment Orchestration & Token Mapping

- **Milestone 1.1:**Establish the baseline development environment. Configure the CMake build system to link the native C++ RDKit runtime environment.

- **Milestone 1.2:**Construct a static dictionary containing localized chemical mapping components. Translate common structural features found in organic systems into robust, syntactically bulletproof SMARTS strings (e.g., ortho/meta/para substitution patterns, aromatic heterocycles, specific alkyl branch structures).
- **Milestone 1.3:**Write a highly streamlined, zero-allocation local CSV parser designed to instantly stream lines of candidate compounds based on mass calculation boundaries.

Phase 2: The Graph Pruning & Geometric Matrix Engine

- **Milestone 2.1:**Implement the core substructure screening loop. Construct verification layers to validate that evaluating a sample pool of 500 candidate compounds completes under 50 milliseconds using native C++ processing.
- **Milestone 2.2:**Build the geometry evaluation module. Write the McConnell anisotropy correction algorithm to shift base predictions based on calculated distances from coordinates generated by RDKit's force-field modeler.
- **Milestone 2.3:**Develop the 2D matrix checker. Design a basic shortest-path graph traversal sequence to score structures based on whether experimental correlations can physically trace a route within valid bond lengths.

Phase 3: Validation, Benchmarking, and Paper Outlining

- **Milestone 3.1:**Ingest genuine historical spectral datasets (e.g., actual hidden configuration challenges provided by university professors) to test true system reliability.
- **Milestone 3.2:**Generate benchmark telemetry contrasting ADSEE's deterministic execution speed and sorting accuracy against traditional increment and machine learning prediction packages.
- **Milestone 3.3:**Draft the final academic manuscript framework, explicitly highlighting how a physics-centric graph constraint solver successfully manages magnetic shielding anomalies that blind generic commercial tools.